

Leveraging Passage Retrieval with Generative Models for Open Domain Question Answering

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Abstract

Generative models for open domain question answering have proven to be competitive, without requiring any explicit knowledge of the target corpus. In this work we investigate how much these models can benefit from retrieval, increasing the number of passages up to 100 while keeping inference time manageable. We propose Fusion-in-Decoder, which processes retrieved passages independently in the encoder and fuses them in the decoder.

1. Introduction

Recent work such as RAG has shown that combining a pre-trained generative model with a dense retriever is an effective way to answer knowledge-intensive questions. We build upon this line of work.

However, RAG encodes retrieved passages jointly with the question, which makes the encoder cost grow linearly with the number of passages. This limits the amount of retrieved evidence a model can actually use.

Unlike RAG, which concatenates passages before encoding, we process each passage independently and defer the fusion to the decoder.

2. Method

Fusion-in-Decoder (FiD) uses a dense passage retriever to fetch the top-k passages for a given question, following the same bi-encoder architecture introduced by DPR.

Each retrieved passage is encoded separately together with the question, and the resulting representations are concatenated and passed to the decoder, which attends to all of them jointly.

This design improves the scaling behaviour with respect to the number of retrieved passages: we can use up to 100 passages at a manageable cost.

3. Experiments

We evaluate on Natural Questions, TriviaQA and SQuAD, where FiD achieves state-of-the-art results among systems that do not use additional pre-training.

Ablation studies show that increasing the number of retrieved passages from 1 to 100 substantially improves exact match accuracy.

We also find that removing the retrieval component causes a large performance drop, confirming that retrieval is the main source of gains.

4. Limitations

Encoding many passages increases memory consumption during training, since all passage representations must be kept for the decoder.

The retriever is still trained separately from the reader, which may leave performance on the table.